Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	[0].25	**1**	
1(b)	$\frac{9}{13}$ cao	**1**	
1(c)	80	**1**	
1(d)	1, 2, 3, 4, 6, 8, 12, 24	**2**	**B1** for 6 or 7 correct factors and no extras or 8 correct factors and at most one extra
1(e)	12	**2**	**B1** for 2, 3, 4 or 6 as final answer or 2 × 2 × 3 or for 2 × 2 × 2 × 3 and 2 × 2 × 3 × 3 × 3
1(f)	Accept any irrational number between 3 and 9	**1**	
1(g)	1	**1**	
1(h)	598.29 cao	**3**	**M2** for $8400\times\left(1+\frac{3.5}{100}\right)^2$ oe OR **M1** for $8400\times\left(1+\frac{3.5}{100}\right)^2$ oe **A1** for 8998.29
1(i)	$\frac{5}{15}$ or $\frac{12}{15}$ or $\frac{35}{15}$	**B1**	allow denominators with multiples of 15 e.g. $\frac{35k}{15k}$, $\frac{5k}{15k}$
	[2] $\frac{5}{15}$ [+] $\frac{12}{15}$ or $\frac{35}{15}$ [+] $\frac{12}{15}$	**M1**	allow other common denominators
	[2] $\frac{17}{15}$ or $\frac{47}{15}$ leading to $3\frac{2}{15}$ cao	**A1**	with no errors or omissions seen
2(a)(i)	3.5	**2**	**B1** for 1, 1, 2, 2, 3, 3, 4 or 3, 4, 4, 4, 5, 5, 7 or 3 and 4 identified
2(a)(ii)	4	**1**	

Question	Answer	Marks	Partial Marks
2(a)(iii)	6	**1**	
2(b)(i)	3.76	**3**	**M2** for $(1 \times 3 + 2 \times 1 + 3 \times 5 + 4 \times 8 + 5 \times 6 + 6 \times 2) \div 25$ or **M1** for $1 \times 3 + 2 \times 1 + 3 \times 5 + 4 \times 8 + 5 \times 6 + 6 \times 2$ or 94
2(b)(ii)	$\frac{16}{25}$ oe	**2**	**B1** for 16
2(c)(i)	136 164 36	**2**	**B1** for one correct
2(c)(ii)	Correct pie chart	**2**	**FT** for 1 or 2 marks provided their 3 angles add to 336° **B1** for one correct sector
3(a)	Reflection $y = -1$	**2**	**B1** for each
3(b)	Rotation [centre] (0, 0) oe 90° [anticlockwise] oe	**3**	**B1** for each
3(c)	Correct triangle vertices (0, −2), (3, −2) and (3, 0)	**2**	**B1** for translation of $\begin{pmatrix} -2 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$
3(d)	Correct triangle vertices (2, 1), (−1, 1) and (−1, −1)	**2**	**B1** for a correct 180° rotation with incorrect centre
4(a)	22 50	**4**	**M1** for 3840 ÷ 720 **A1** for 5[h] 20[m] **M1dep** for 15 30 + *their* 5[h] 20[m] [+2] or 17 30 + *their* 5[h] 20[m]
4(b)	75	**2**	**M1** for 2610 ÷ 34.8
4(c)	6.5	**3**	**M2** for $\frac{19.17 - 18}{18}$ [× 100] oe or $\left(\frac{19.17}{18} - 1\right)$ [× 100] oe or $\frac{19.17}{18} \times 100$ [– 100] oe or **M1** for $\frac{19.17}{18}$ or 19.17 – 18